Tallinn University of Technology

ATR0030 Robotics

Lab 2. Virtual robotic station with ABB RobotStudio

Drawing a circle and the initials “JB” with an ABB IRB 1600 robot

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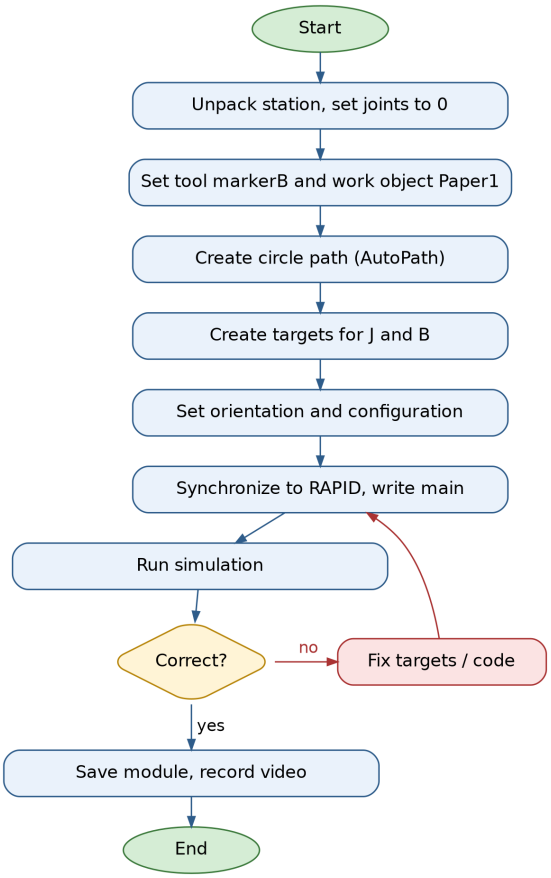
Task

The task of this work was to prepare a program in ABB RobotStudio for an ABB IRB 1600 robot arm to draw a circle and initials (JB) on a sheet of paper, and to prepare this program for execution in Lab 3, where the real robot will draw the figure. A virtual robotic station was configured, and the robot was taught using two methods of target design: binding targets to a pre-defined geometry (the circle, created with AutoPath around a cylinder) and free indication of targets (the initials J and B). The paths were programmed with movement instructions in the RAPID language and tested in simulation.

a. Robot name, degrees of freedom, body names and joint types

Robot name	ABB IRB 1600-6/1.45 (`IRB1600_6_145__02`)
Degrees of freedom	6
Body names	Base, Link1, Link2, Link3, Link4, Link5, Link6
Joint types	6 revolute joints

b. Flowchart of the program development process



c. Executable RAPID program with comments

```
MODULE Module1
    ! ATR0030 Lab 2 - Jordan Brodie
    ! Draws a circle (Path_10) and the initials J (Path_20) and B (Path_30) on Paper1 with markerB.
    ! Target orientation [0,0.707,0.707,0] = marker pointing straight down; config [-1,-1,0,1] = Cfg3.
    ! Z = 0 -> pen on paper, Z = 10 -> pen lifted (UP targets, step 2.4.5)

    ! --- Circle targets (AutoPath around cylinder r = 70 mm, centre 148.5,105) ---

    CONST robtarget Target_10:=[[148.5,35,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_20:=[[218.5,105,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_30:=[[148.5,175,0.1],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_40:=[[78.5,105,0.1],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_50:=[[148.5,35,0.1],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];

    ! --- Letter J targets (free targets, method 2) ---
    CONST robtarget Target_60:=[[96.373,122.697,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_60up:=[[96.373,122.697,10],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_70:=[[128.291,123.702,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_80:=[[127.45,85.334,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_90:=[[99.327,88.197,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];



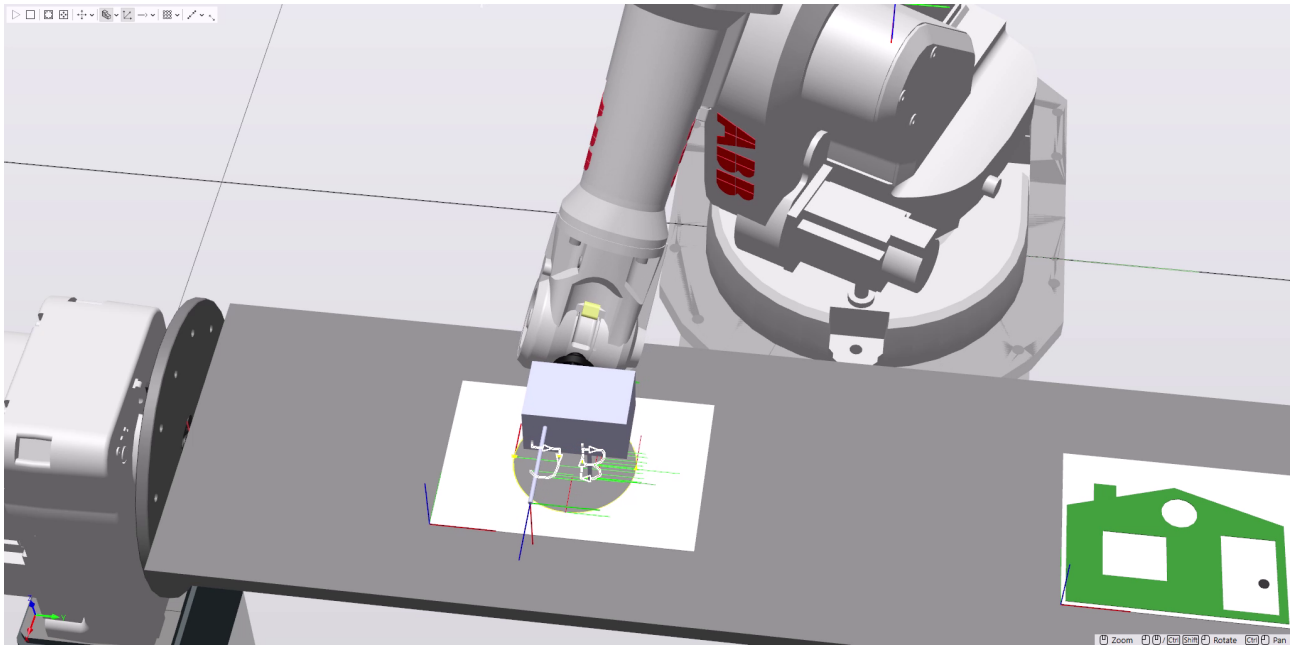
    ! --- Letter B targets (free targets, method 2) ---
    CONST robtarget Target_100:=[[153.65,126.951,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_100up:=[[153.65,126.951,10],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_110:=[[176.968,128.117,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_120:=[[169.847,106.303,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_140:=[[169.849,106.04,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_130:=[[156.283,105.762,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_150:=[[172.387,86.337,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_160:=[[158.109,85.033,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_170:=[[154.681,128.259,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    ! --- Extra points added to shape J (220-240) and B (250, 180-210) ---
    CONST robtarget Target_220:=[[129.862,99.321,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_230:=[[120.736,81.348,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_240:=[[107.716,81.075,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_250:=[[179.813,123.004,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_180:=[[180.068,115.515,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_190:=[[176.304,110.207,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_200:=[[177.546,99.997,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_210:=[[179.316,92.92,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];

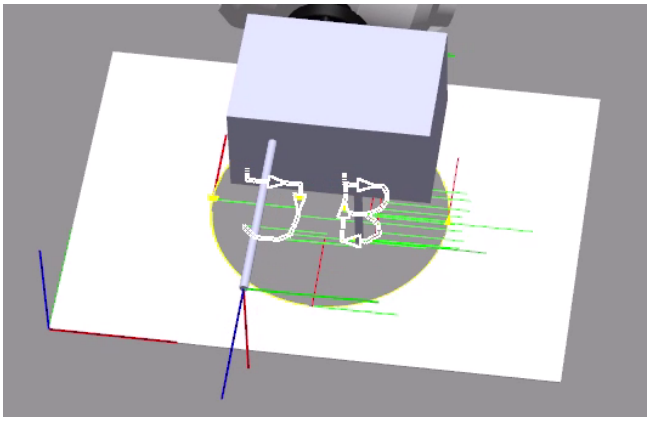

    PROC main() ! Jordan Brodie entry point
Path_10;   ! draw the circle
Path_20;   ! draw the letter J
```

```
Path_30;   ! draw the letter B
MoveAbsJ calib_pos, v1000, z0, tool0;   ! return to calibration position (all joints 0)
Stop;   ! end of program
ENDPROC


    PROC Path_10() ! circle: two circular moves through four points
        MoveJ Target_10,v1000,z100,markerB\WObj:=Paper1;
        MoveC Target_20,Target_30,v1000,z100,markerB\WObj:=Paper1;
        MoveC Target_40,Target_50,v1000,z100,markerB\WObj:=Paper1;
    ENDPROC
    PROC Path_20() ! letter J
        MoveJ Target_60up,v1000,z0,markerB\WObj:=Paper1;   ! hover 10 mm above start of J
        MoveJ Target_60,v1000,z0,markerB\WObj:=Paper1;     ! pen down
        MoveJ Target_70,v1000,z0,markerB\WObj:=Paper1;     ! top bar
        MoveJ Target_220,v1000,z0,markerB\WObj:=Paper1;    ! stem
        MoveJ Target_80,v1000,z0,markerB\WObj:=Paper1;
        MoveJ Target_230,v1000,z0,markerB\WObj:=Paper1;    ! hook
        MoveJ Target_240,v1000,z0,markerB\WObj:=Paper1;
        MoveJ Target_90,v1000,z0,markerB\WObj:=Paper1;
    ENDPROC
    PROC Path_30() ! letter B
        MoveJ Target_100up,v1000,z0,markerB\WObj:=Paper1;  ! hover 10 mm above start of B
        MoveJ Target_100,v1000,z0,markerB\WObj:=Paper1;    ! pen down
        MoveJ Target_110,v1000,z0,markerB\WObj:=Paper1;    ! top
        MoveJ Target_250,v1000,z0,markerB\WObj:=Paper1;    ! upper bowl
        MoveJ Target_180,v1000,z0,markerB\WObj:=Paper1;
        MoveJ Target_190,v1000,z0,markerB\WObj:=Paper1;
        MoveJ Target_120,v1000,z0,markerB\WObj:=Paper1;
        MoveJ Target_140,v1000,z0,markerB\WObj:=Paper1;
        MoveJ Target_130,v1000,z0,markerB\WObj:=Paper1;    ! middle bar to stem
        MoveJ Target_140,v1000,z0,markerB\WObj:=Paper1;
        MoveJ Target_200,v1000,z0,markerB\WObj:=Paper1;    ! lower bowl
        MoveJ Target_210,v1000,z0,markerB\WObj:=Paper1;
        MoveJ Target_150,v1000,z0,markerB\WObj:=Paper1;
        MoveJ Target_160,v1000,z0,markerB\WObj:=Paper1;    ! bottom
        MoveJ Target_170,v1000,z0,markerB\WObj:=Paper1;    ! stem back up, closes the B
    ENDPROC
ENDMODULE
```

d. Screenshot of the paths created in the graphics area





e. Conclusions

In this lab I learned more about how industrial robots work and how they are programmed. The process was similar to the Mitsubishi robot in Lab 1: first the points (targets) are defined, and then a path is generated that moves the tool through those points in order. In RobotStudio this was done in a virtual station, where the station was configured by unpacking it, setting all joints to 0 to match the real robot's calibration position, and choosing `markerB` as the tool and `Paper1` as the work object.

The software was confusing at first, especially the steps for orientation, configuration and synchronising between the station and the RAPID code. Once I understood how these parts fit together, it became much easier to work with. The circle was quick to make with AutoPath, but the initials took more work. To get the shape of the B that I wanted, I had to add extra targets for the curves and reuse some of the targets more than once in the path (for example `Target_140` for the middle bar). I also copied the first target of each letter and raised it by 10 mm to make the UP targets, so the marker lifts off the paper between the shapes instead of drawing a line across the page.

The program ran correctly in simulation and drew the circle and the initials JB, so it is ready to be tested on the real robot in Lab 3. Before then, the speed should be reduced and linear moves used for drawing so that the lines on the paper come out clean.

This lab inspired me to work more with robots and to keep learning how RobotStudio works, since it makes it possible to plan and test a robot program safely before running it on real hardware.

f. Video of the program execution

The simulation of the program execution was recorded with RobotStudio and is submitted as a separate file: **JordanBrodieLab2VideofProgram.mp4**. Frames from the video are shown below.

